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Racetrack benchmark (2-D)

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The first benchmark presented is a 2-D tape stack, representing an infinitely long racetrack coil. The stack contains 19 REBCO tapes. The width of the tapes is 4 mm, while their thickness is 1 µm. The spacing between the tapes is 250 µm. The geometry and the mesh used for the simulation are presented in Fig. 1.

The resistivity of the REBCO tapes is computed using the power-law from the equation

$$\rho = \frac{E_c}{J_c}\left(\frac{|\mathbf{J}|}{J_c}\right)^{n-1}, \tag{1}$$

using the critical current density $J_c(B,\theta)$ and the $n(B,\theta)$ factor from the SuperPower Advanced Pinning data in the Robinson Database [1] at 77 K. The spacing between the tapes is made of copper [2], thus electrically connecting all the tapes. This also implies that there is only a single 1-cohomology cut to impose the total current that is distributed between all the tapes and the copper, as shown in Fig. 1. The simulation is driven by an AC current. The frequency of the sine wave is $f = 50\,\mathrm{Hz}$ and the amplitude is 160 A per tape[1]. Each tape is modelled with the thin-shell model, using one element across the thickness. The condition considered at the outer air boundary is a perfect electric conductor boundary condition ($\mathbf{n}\cdot\mathbf{B} = 0$), which we recall arises naturally in the ϕ formulation's weak form and is automatically enforced if no other condition is applied. In terms of the BELFEM features described in this thesis, this benchmark highlights the automatic cohomology computation, the thin-shell model and the **H**-**H** coupling for current sharing.

Fig. 2 presents the distribution of the magnetic flux density norm at $t = T/4$, $T/2$, $3T/4$ and T, where T is the period of the sine wave. The effect of the permanent currents and the field penetration within the stack of superconducting tapes can be observed at $t = T/2$ and $t = T$, when the total transport current is 0 A. Similarly, Fig. 3 presents the current density distribution along the x axis in the middle tape at $t = T/4$ and $T/2$, displaying the current saturation around $J = J_c$. We also note that, in this simulation, the critical current density depends on the flux density norm and angle with the tape's plane, making it

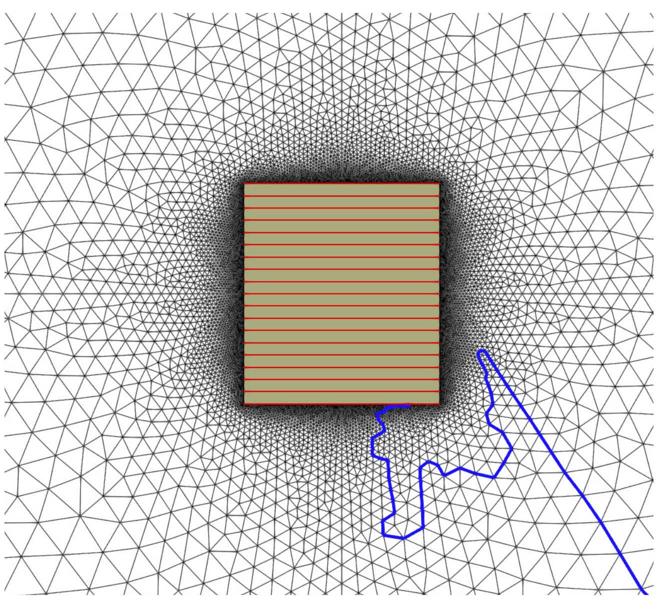

Figure 1: 2-D racetrack geometry and mesh consisting of 19 REBCO tapes (in red), separated by copper. The thin cut is represented in blue.

[1] Since only one cut is defined, the total current of 3040 A is imposed in the simulation, thus allowing an uneven distribution of the current between the tapes, depending on the underlying physics.

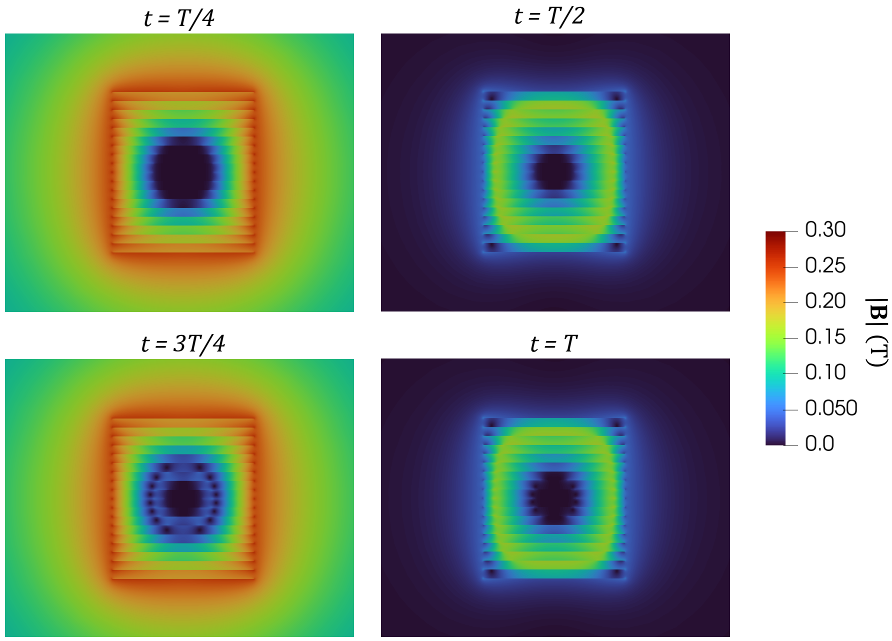

Figure 2: Magnetic flux density distributions at various times ($T/4$, $T/2$, $3T/4$ and T, where T is the period of the signal).

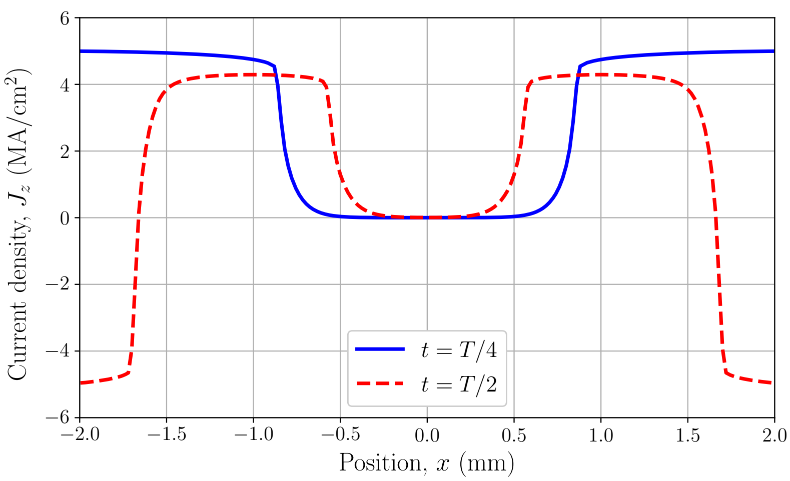

Figure 3: Distribution of the current density along the x axis of the middle tape at $t = T/4$ and $t = T/2$, where T is the period of the signal

non-uniform along the tapes and across the stack, allowing some tapes to carry more current than others. For example, as shown in Fig. 2, the flux density is maximal on the top and bottom tapes at $t = T/4$, which effectively decreases J_c on those tapes compared to the middle one. Fig. 3 also displays this effect with the variation of the saturation current locally and as a function of time during the simulation, depending on the amplitude of the flux density.

1 References

[1] S. C. Wimbush and N. M. Strickland, "A public database of high-temperature superconductor critical current data," *IEEE Transactions on Applied Superconductivity*, vol. 27, no. 4, pp. 1–5, 2016.

[2] J. G. Hust and A. B. Lankford, "Thermal conductivity of aluminum, copper, iron, and tungsten for temperatures from 1 k to the melting point," tech. rep., National Bureau of Standards, Boulder, CO (USA). Chemical Engineering . . . , 1984.